Bike-Sharing Demand: Should We Just Buy More Bikes?

Case study report: data quality and exploration, a tested hypothesis, a fleet-sizing model, and an AI assistant that lets operations ask about demand in plain English.

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Data	UCI Bike Sharing dataset: Capital Bikeshare, Washington D.C., 2011-2012 (731 days, 17,379 hours)
Code and notebooks	https://github.com/anuragparasar/Bike-Sharing-Company-Case-Study

Executive summary

The CEO's question: riders can't find a bike when they need one, yet bikes sit unused all over the city. Should we just buy more bikes?

Short answer: not before checking two things. Both complaints are true at the same time because demand is concentrated in a few hours. A bike is on a trip only about **8%** of the time, and the fleet is only tested in about **7%** of hours: working-day evenings (17:00-18:00) and weekend afternoons.

- **Demand is peaky, and the peaks are predictable.** On working days the 3 busiest hours carry 33.4% of the day's rentals against 27.4% on weekends ($p < 10^{-70}$); at the same daily volume the working-day peak hour is 28.5% higher.
- **The model is accurate enough to plan with.** On four unseen months it explains 91% of hourly variation (R^2 0.91) against 0.68 for a simple baseline, and cuts the error at working-day peaks from 131 to 84 rentals per hour.
- **About 720 bikes cover 95% of days** (plausible range 600-1,100, depending on trip length and how many parked bikes riders need nearby). Covering 95% of days instead of 80% costs only about 50 extra bikes.
- **Recommendation:** compare the real fleet with about 720. If the fleet is already larger, shortages come from *where* bikes are, not how many, so rebalance before the peaks rather than buy. Run the rebalancing pilot described in section 3 before any large purchase, and collect station-level data.
- **Prototype:** an assistant answers questions such as "How many bikes do we need next Friday at 6pm if it rains?" Every number comes from the model; a LangGraph agent with Gemini handles free-form language.

1. The data and its quality

The dataset records rentals from Capital Bikeshare (Washington D.C.) for 2011-2012, aggregated per day and per hour, with weather and calendar information added. Each row gives rentals by **registered members** and **casual riders**; their sum is total rentals. Weather values are pre-scaled to 0-1 and were converted back to °C, % humidity and km/h.

Quality checks

Check	Finding	How it was handled
Missing values, duplicates	None. casual + registered = total on every row.	-
Missing hours	165 of 17,544 hours absent: mostly 02:00-05:00 (no rentals), plus outages such as Hurricane Sandy (29-30 Oct 2012).	Treated as zero-rental hours; outage days excluded from the hypothesis test.
Wind speed = 0	2,180 hours: almost certainly missing readings, not calm air.	Set to missing; the model handles missing values natively.
Humidity = 0	22 hours: physically impossible, a sensor gap.	Set to missing.
Season labels	**The readme is wrong.** Code 1 covers 21 Dec-20 Mar, i.e. winter, not spring.	Relabelled: 1 = winter, 2 = spring, 3 = summer, 4 = fall.
Real-world anomalies	Hurricane Sandy (22 rentals in a day), low days around Christmas.	Kept for training; flagged in evaluation.

What the data cannot show

- It is **city-wide totals per hour**: no stations, no bike locations, no fleet size.
- It records **rentals that happened**, not riders who found no bike. True peak demand is therefore *higher* than recorded (censored), so every peak estimate here is a floor.
- So the data can show *when* demand peaks, but not *where* bikes run out. That requires station-level data.

2. What demand looks like

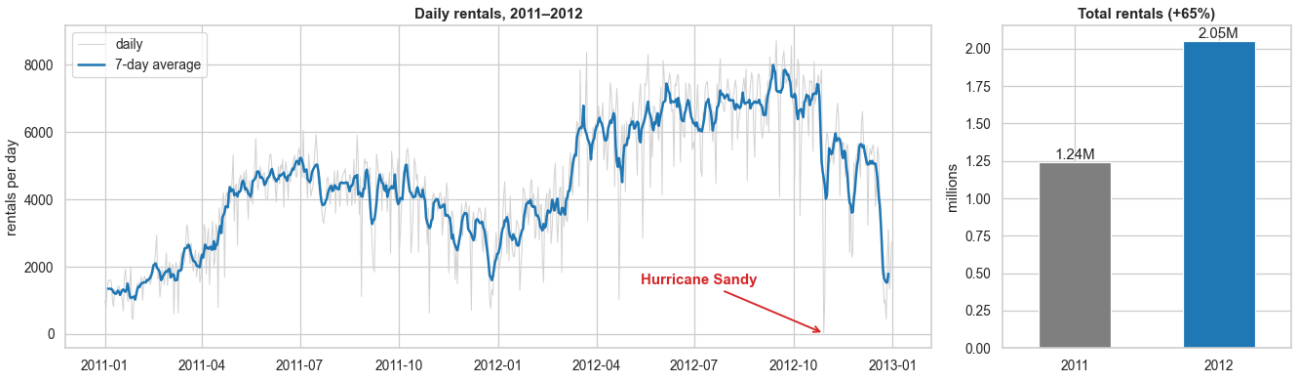


Figure 1. Daily rentals grew 65% from 2011 (1.24M) to 2012 (2.05M), with a strong seasonal cycle. Hurricane Sandy shows as a one-day collapse.

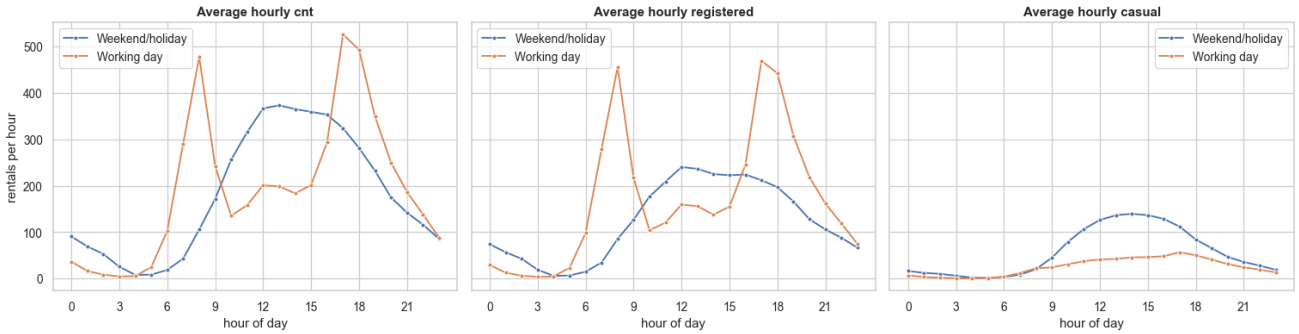


Figure 2. Two different daily rhythms. Working days have sharp commute peaks at 08:00 and 17:00-18:00, driven by members; weekends have a broad midday hump with more casual riders.

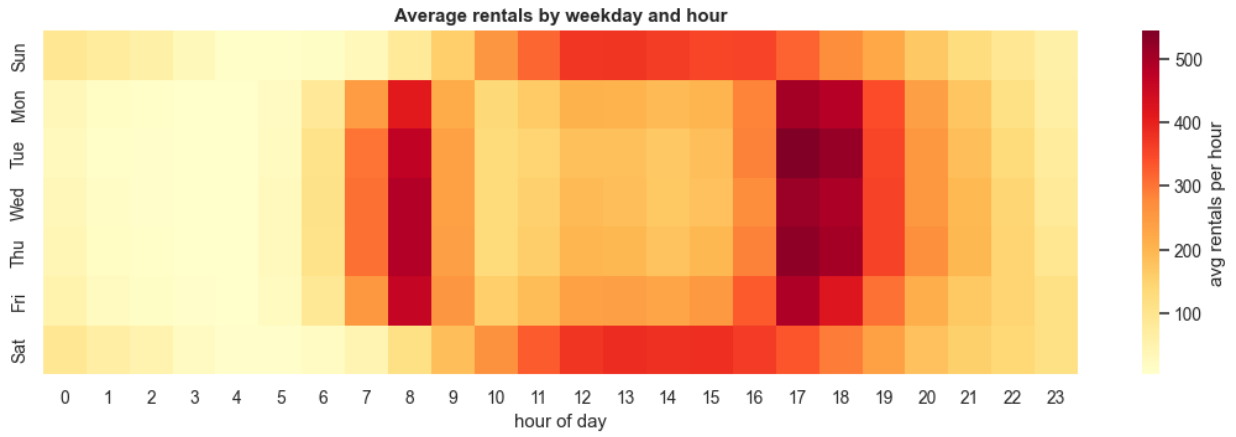


Figure 3. Average rentals by weekday and hour: the commute peaks are the hottest cells of the week.

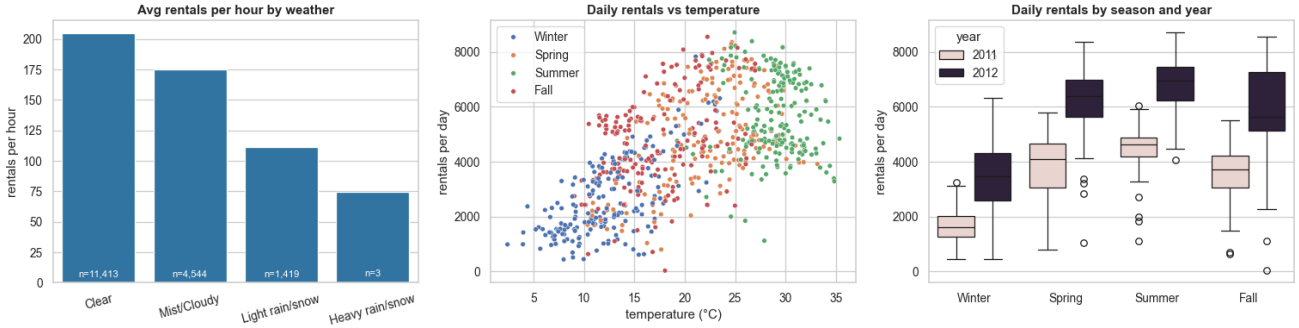


Figure 4. Bad weather cuts demand sharply (light rain or snow roughly halves it); demand rises with temperature and is higher in every season of 2012 than 2011.

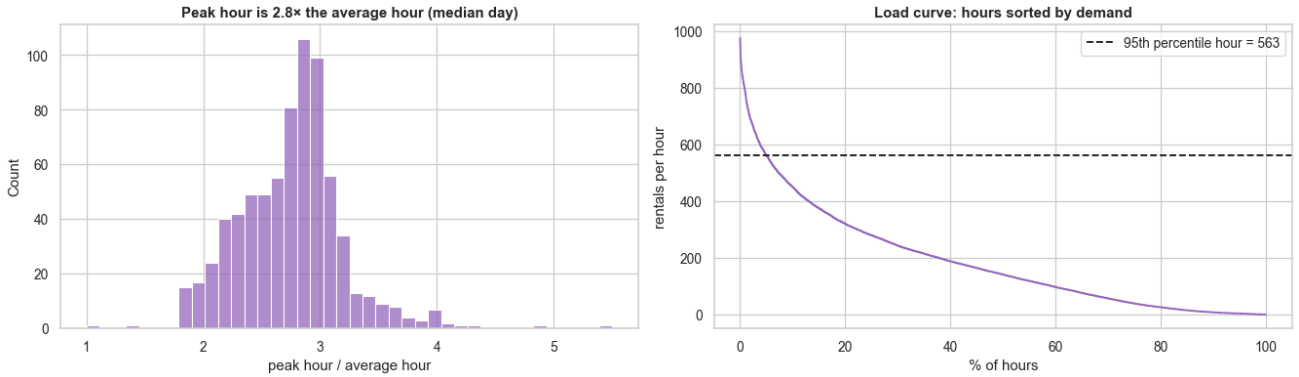


Figure 5. On a typical day the busiest hour is 2.8 times the average hour, and only 25% of hours reach even half of the 95th-percentile demand. A fleet sized for the peak is idle most of the time.

3. A testable hypothesis

If demand is concentrated into a few hours, the fleet must be sized for those hours and sits idle the rest of the day, which explains both complaints. The sharpest version this data can test is about commuting:

> **H_0:** the share of a day's rentals in its 3 busiest hours is the same on working days and non-working days.
>
> **H_1:** the share is higher on working days, i.e. working-day demand is more concentrated in time.

The top-3-hour share measures concentration independently of how busy a day is, matches a commute window, and links directly to fleet sizing (the fleet follows the peak; utilisation follows the total). Eight outage days were excluded, leaving 723 days.

Test	Result
Median top-3-hour share	**33.4%** on working days vs **27.4%** on weekends and holidays (an even spread would be 12.5%)
Difference (bootstrap 95% CI)	+6.0 points (+5.2 to +6.7)
Mann-Whitney U, one-sided	p about 2 x 10^{-72}; rank-biserial effect size 0.83 (very large)
Permutation test (10,000 shuffles)	No shuffle reached the observed gap
Regression controlling for daily volume, weather, season, year	Working-day peak hour **+28.5%** (95% CI +26.5% to +30.6%), R^2 0.96
Robustness	Holds in all 8 season x year groups (+3.4 to +7.3 points, every $p < 10^{-5}$)

Result: H_0 is rejected. Working-day demand is far more concentrated, and the effect is not explained by volume, weather, season or year.

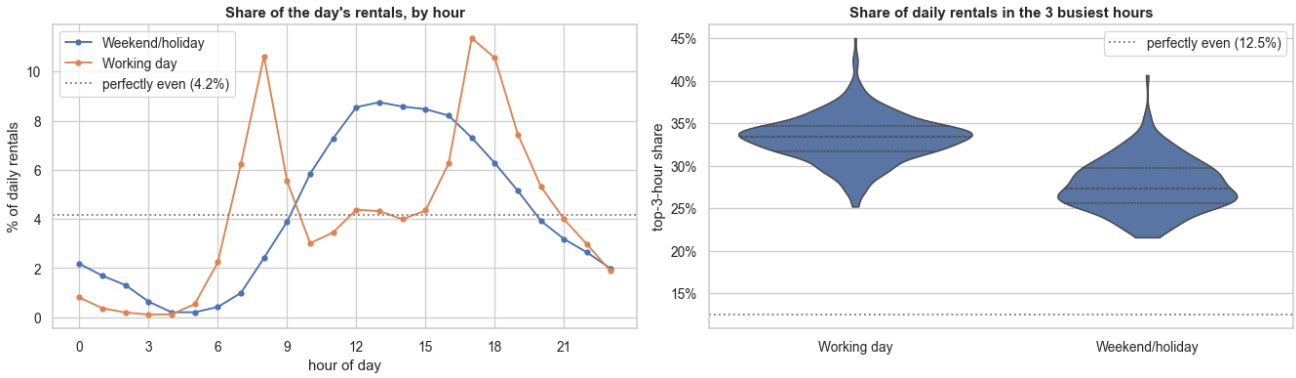


Figure 6. Left: share of the day's rentals by hour. Right: the top-3-hour share is clearly higher on working days.

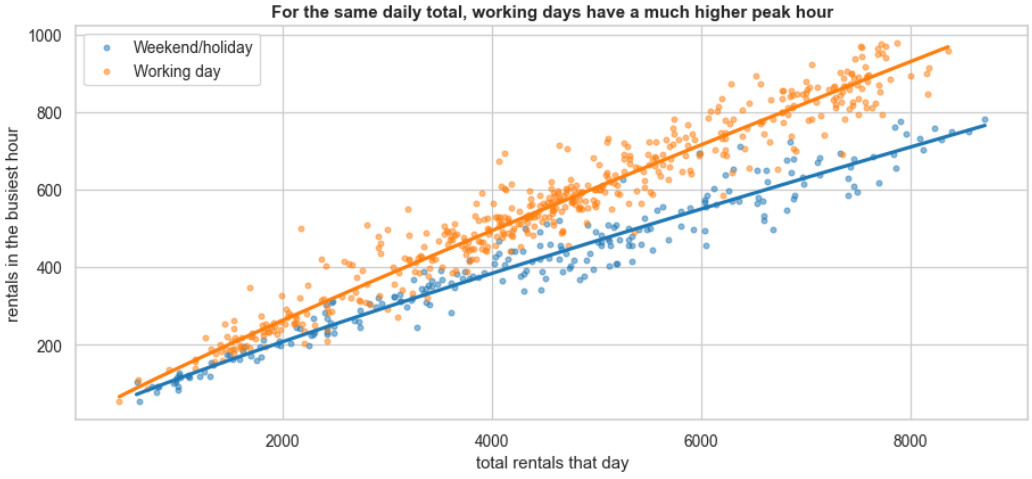


Figure 7. For the same daily total, working days have a much higher busiest hour.

How to test the real decision in the field

The data supports "demand is concentrated in time". The CEO's decision (buy bikes or manage the peaks) needs a controlled experiment with station-level data:

Element	Design
Hypothesis	Targeted rebalancing before the commute peaks reduces stockouts at least as much as adding 10% more bikes, at lower cost.
Unit and arms	Station zones, randomised (stratified by zone type and baseline demand) into: control; pre-peak rebalancing; +10% bikes.
Primary metric	Empty-station minutes during peak hours (07-10, 16-19) per zone-day.
Secondary and guardrails	Rides per bike per day, failed unlock attempts, cost per avoided stockout; full-dock minutes, complaints.
Duration and analysis	2-week baseline + 6 weeks; difference-in-differences with zone fixed effects, errors clustered by zone.
Size	About 64 zones per arm for a moderate effect (Cohen's d 0.5, 80% power); a week-by-week switchback design if zones are few.
Decision rule	If rebalancing cuts stockouts as much as extra bikes at lower cost, rebalance instead of buying.

4. How many bikes are needed?

The model works in three steps: forecast hourly rentals, convert rentals into bikes on the road at the same time, and size the fleet for the busy hours.

- **Forecast:** gradient-boosted trees with a Poisson loss (suited to counts), one model for members and one for casual riders, using hour, day type, weekday, holiday, season, month, year and weather. Lagged demand was left out on purpose so the model can answer what-if questions such as "next Friday if it rains".
- **Validation:** trained on Jan 2011-Aug 2012 and tested on Sep-Dec 2012, which the model never saw.
- **Bikes in use:** rentals x trip length (Little's law), with members assumed to ride 12 minutes and casual riders 30 minutes.
- **Safety margin:** a 90th-percentile model of bikes in use, calibrated on held-out July-August data so that demand stays under the plan 9 hours out of 10.
- **Fleet:** cover the busiest hour on 95% of 2012 days, plus 1.5 parked bikes per bike in use (so riders find one nearby) and 10% in repair.

Accuracy on the four unseen months

Model	Average error (rentals/hour)	Relative error (RMSLE)	R^2	Error at working-day peaks
Baseline: recent average for the same hour, day type and weather	77.3	0.638	0.680	130.8
Gradient boosting, total rentals	41.9	0.372	0.909	84.4
Gradient boosting, members + casual (used)	**42.3**	**0.377**	**0.911**	**83.9**

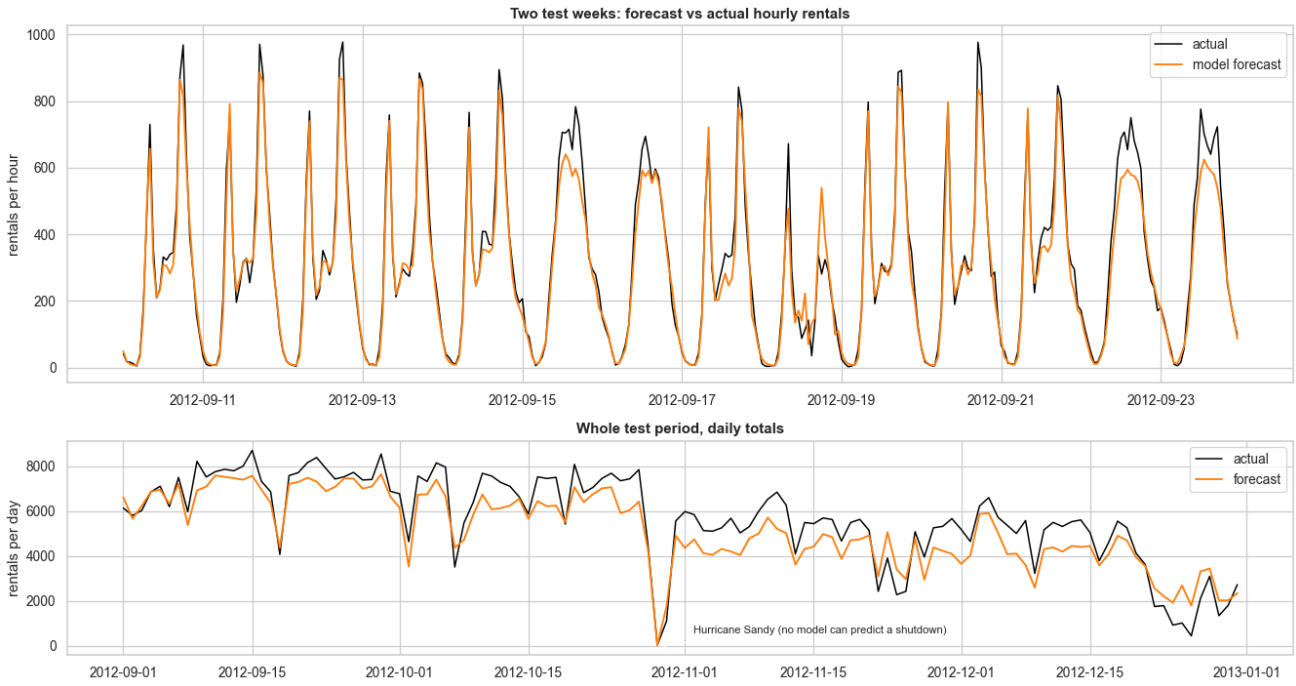


Figure 8. Forecast vs actual on unseen data. Top: two test weeks. Bottom: daily totals; the model runs 10-15% low in Nov-Dec because demand kept growing beyond the training data.

The planning margin needed calibration: before it, the 90% plan covered only 76% of unseen hours (55% at working-day peaks) because of that growth. Scaling it by 1.126, learned on July-August, brought coverage to **91%** overall and **89%** at peaks, matching the 90% target.

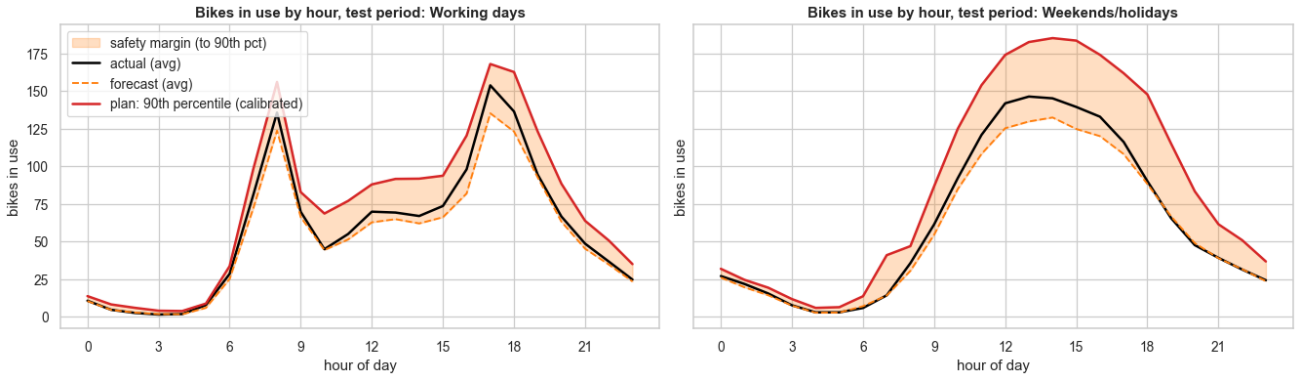


Figure 9. Bikes in use by hour in the test period. In bikes-in-use terms, weekend afternoons are as demanding as weekday evenings because leisure riders keep bikes about 2.5 times longer.

Fleet size

Result	Value
Design peak: bikes on the road at once (95% of 2012 days)	about 259
Total fleet needed (with parked bikes and repairs)	**about 720**
Average bikes on trips across 2012	60, so each bike is in use about 8% of the time
Hours near capacity (above 80% of the design peak)	7% of 2012 hours: 43% of them working-day 17:00-18:00, 52% weekend afternoons 11:00-17:00

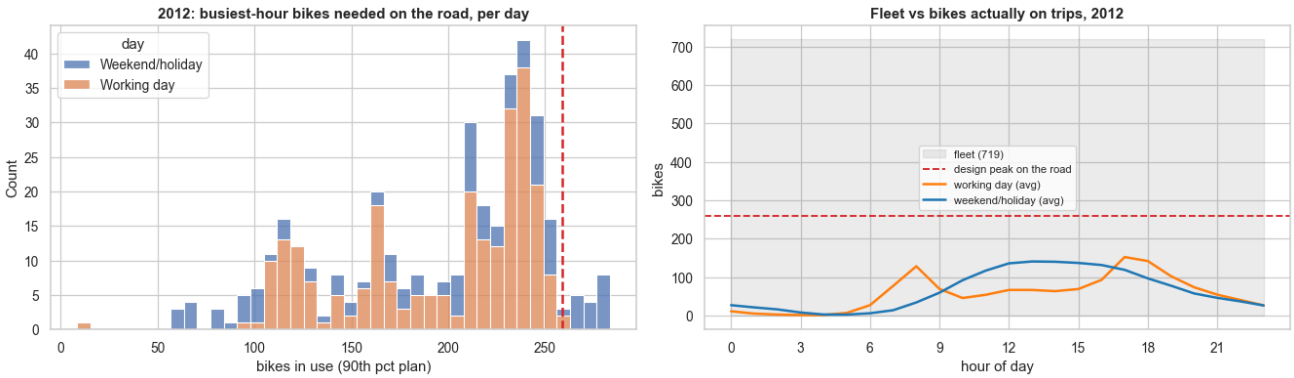


Figure 10. Left: busiest-hour requirement per day in 2012, with the design peak. Right: the fleet against bikes actually on trips on an average day.

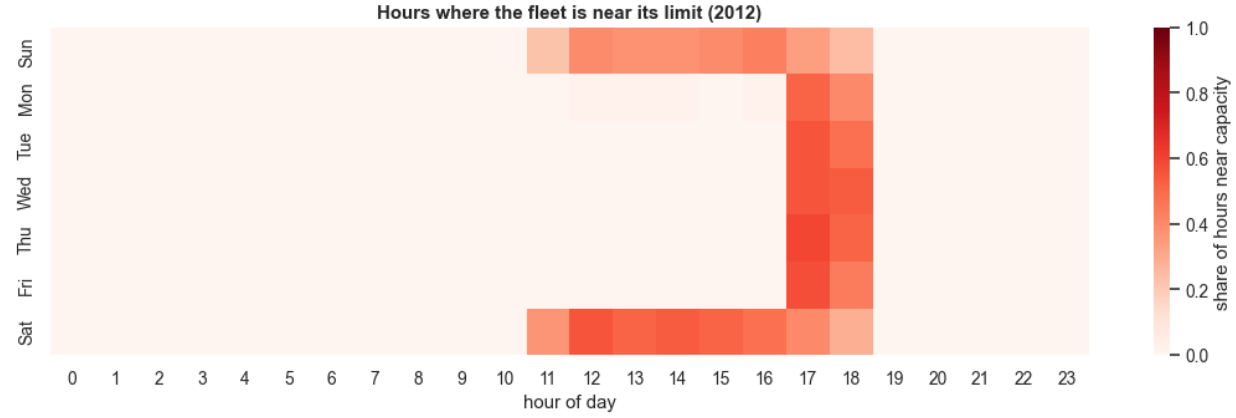


Figure 11. The only hours where the fleet is tested: working-day evenings and weekend afternoons.

Trade-off and sensitivity

Share of days covered	Bikes on the road	Total fleet	Days over capacity per year	of which weekends/holidays
80%	241	670	73	37
90%	249	691	37	26
95% (recommended)	**259**	**719**	**19**	**18**
99%	279	776	4	4
100%	283	787	0	0

Trip lengths and the parked-bike buffer are not in the data, so the fleet was recomputed across plausible values:

Trip length (member / casual)	Buffer 1.0	Buffer 1.5	Buffer 2.0	Buffer 2.5
10 / 25 min	480	600	720	840
12 / 30 min (base case)	580	**720**	860	1,010
15 / 35 min	700	870	1,050	1,220
20 / 40 min	880	1,100	1,330	1,550

The answer moves from about 600 to about 1,100 bikes with these assumptions, so measuring trip lengths and station-level availability is the single most valuable next step.

5. Prototype: asking about demand in plain English

The operations manager plans staff, rebalancing trucks and maintenance, and wants answers, not notebooks. The prototype is a chat app (Streamlit) backed by three tools that call the model:

Question type	Example	Tool
Plan a specific day	How many bikes do we need next Friday at 6pm if it rains?	forecast_day
What-if	How much does rain change demand on 14 June compared to clear weather?	forecast_day (twice)
Understand patterns	What are the busiest hours on weekends? How does weather affect rentals?	historical_demand
Strategic	Do we need to buy more bikes?	fleet_summary

Design principles

- **Every number comes from a tool, never from the language model.** The language layer only interprets the question and phrases the answer, which removes the main risk of AI assistants: confident made-up numbers.
- **Show the working.** Each answer comes with an hourly chart against fleet capacity, the data table behind it, and the exact tool calls made.
- **State the assumptions that matter:** 2012 demand levels unless a growth % is given, the 90% planning level, typical monthly temperature when none is given, and that heavy rain is rare in the training data. US federal holidays are detected automatically.
- **Works without an API key.** A rule-based parser handles common phrasings offline; the AI agent is an upgrade.

Two language layers

	Offline (rule-based)	AI agent (LangGraph + Gemini)
How it understands questions	Patterns for dates, weather, times, temperatures and growth	Gemini (via LangChain) reads the question and chooses tools and arguments
Follow-ups ("and if it's sunny?")	No	Yes: a LangGraph checkpointer keeps each conversation's history
Errors	Unrecognised questions get a help message	Bad tool arguments are returned to the model, which corrects itself; the loop is capped at 12 steps
Needs	Nothing	GOOGLE_API_KEY; falls back to offline if unavailable

The agent is a LangGraph state graph: an **agent** node (Gemini with the tools attached) either requests tools or answers; a **tools** node runs the requested tools locally, in parallel if needed, and returns the results; the graph loops until the model answers. It was tested end to end with a scripted stand-in model (parallel tool calls, error recovery, memory across turns); it has not yet been run against live Gemini.

How well does the offline parser understand questions?

On 19 standard phrasings written after the parser, it got **19 of 19** right. On 5 harder, more natural ones ("noon", "the 5th", a follow-up, "Easter weekend", "evening crunch in winter") it got **0 of 5**. That is the case for the AI agent; a proper evaluation would use real questions logged from the operations team.

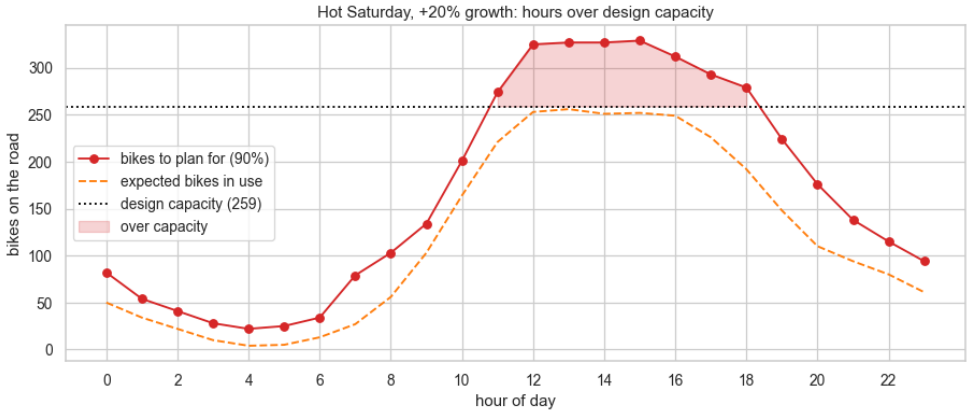


Figure 12. What the manager sees for "Saturday at 2pm, 30 degrees and sunny, with 20% growth": the plan exceeds the fleet's design capacity from 11:00 to 18:00, and the assistant flags it.

6. Recommendation

> **Don't buy bikes across the board.** Manage the peaks first, measure, then buy only for the gap that remains.

Step	Action	Why
1	Compare the current fleet with about 720 bikes (range 600-1,100).	If the fleet is already larger, shortages cannot be a city-wide fleet-size problem: they are about where bikes are.
2	Rebalance before 16:00 on working days toward commute origins.	Working-day evening shortages are a distribution problem.
3	Use trip time limits or pricing on weekend afternoons, and supply near parks and leisure areas.	Weekend pressure comes from long leisure trips tying up bikes.
4	Run the rebalancing pilot (section 3) before any large purchase.	It answers "manage or buy" directly, with costs.
5	Collect station-level data: dock status every few minutes, trip origins, destinations and durations.	Replaces the two biggest assumptions with measurements and shows where bikes run out.
6	Buy only the remaining gap, sized with the model plus a growth assumption.	With +20% demand the base-case fleet becomes about 860 bikes.

Limitations

- City-wide data cannot see spatial imbalance, the most likely cause of "can't find a bike".
- Recorded rentals understate true peak demand, so the design peak is a floor.
- Trip lengths and the parked-bike buffer are assumptions; the fleet estimate ranges from about 600 to 1,100 with them.
- The model has seen only 2011-2012; future growth is an explicit input, not learned.
- The AI agent's tool calling was tested with a scripted stand-in model, not yet with live Gemini.

Repository

All code, notebooks and the prototype: https://github.com/anuragparasar/Bike-Sharing-Company-Case-Study

File	Contents
note.ipynb	Part 1: dataset explanation, quality checks, feature preparation, exploratory charts
2_hypothesis.ipynb	Part 2: hypothesis, four statistical tests, field experiment design with power calculation
3_model.ipynb	Part 3: demand model, bikes-in-use conversion, calibrated planning margin, fleet sizing, sensitivity
4_prototype.ipynb	Part 4: assistant design, demo answers, parser evaluation, LangGraph agent test
bike_model.py	Shared features and fleet-sizing assumptions
demand_assistant.py	Tools, offline parser and the LangGraph + Gemini agent
app.py	Streamlit chat app (run: streamlit run app.py)
models/bike_demand.joblib	Trained models and design numbers used by the prototype